

Reporte de ejecución - SecureBank

Fecha de ejecución: 2026-09-23T10:32:09.406169-06:00. Plataforma: Windows, Python 3.12.14, pytest 9.1.1, pytest-cov 7.1.0. Reloj del sistema bajo prueba: 2026-09-01 UTC, controlado por fixture.

Resumen

Métrica	Resultado
Total	125
Aprobados	125
Fallidos / errores / omitidos	0 / 0 / 0
Duración JUnit	0.716 s
Líneas	150/150 (100.00%)
Ramas	52/52 (100.00%)
Funciones	No se reporta como porcentaje: pytest-cov no genera esta métrica en el resumen utilizado

Comando reproducible:

```
python -m pytest -v --cov=src --cov-branch --cov-report=term-missing --cov-report=json:reports/coverage.json --cov-report=html --junitxml=reports/results.xml
```

El reporte de terminal y JUnit pueden diferir ligeramente en duración porque miden intervalos distintos. Pylint: ver [salida íntegra](#). Evidencia primaria: [pytest](#), [JUnit](#), [coverage JSON](#).

Cobertura por técnica, ejecutada aisladamente

Técnica	Casos	Líneas	Ramas	Combinada
equivalence_partitioning	38	113/150 (75.3%)	33/52 (63.5%)	72.3%
boundary_values	37	94/150 (62.7%)	23/52 (44.2%)	57.9%
decision_tables	38	120/150 (80.0%)	32/52 (61.5%)	75.2%
state_transitions	12	103/150 (68.7%)	24/52 (46.2%)	62.9%

Las mediciones individuales se obtuvieron con `python -m pytest tests/test_<técnica>.py --cov=src --cov-branch --cov-report=json`. Los cuatro JSON individuales están junto al reporte. No se suman porcentajes: existe solapamiento.

Experimento de detección de defectos

La implementación inicial pasó los 125 casos. No se inventan defectos encontrados durante desarrollo. Para comparar sensibilidad se introdujeron cinco mutaciones deliberadas, una a la vez en copias temporales de `src/` y `tests/`, ejecutando cada técnica de forma aislada. Exit code 1 significa que al menos una aserción detectó el cambio; errores de colección no se contabilizan como detección. Se restauró siempre el código original y se repitió la suite final. Este experimento pequeño no constituye una tasa general de detección.

Mutación	Cambio	EP	BVA	DT	ST
M1	Accept zero transfers	Detectado	Detectado	Detectado	No detectado
M2	Reject an exact daily limit	No detectado	Detectado	Detectado	No detectado

Mutación	Cambio	EP	BVA	DT	ST
M3	Waive fees at the exact threshold	No detectado	Detectado	Detectado	No detectado
M4	Allow frozen outgoing transactions	No detectado	No detectado	Detectado	Detectado
M5	Force Suspended after every ledger movement	Detectado	Detectado	Detectado	Detectado

Las sustituciones exactas están en [mutations.json](#); M5 se aplica exclusivamente a la recalculación de estado en `_record`, no al constructor. Los veinte registros `mutation-M*-<técnica>.txt` incluyen fallos y aserciones. Para reproducir, copiar el proyecto a un directorio temporal, aplicar una sola sustitución y ejecutar los cuatro archivos por separado. No aplicar mutaciones al archivo de entrega.

Capturas y validación

```
pytest: tramo final del registro real

tests/test_equivalence_partitioning.py::test_update_information_partition[EP-I2] PASSED [ 86%]
tests/test_equivalence_partitioning.py::test_update_information_partition[EP-I3] PASSED [ 87%]
tests/test_equivalence_partitioning.py::test_transfer_between_own_accounts PASSED [ 88%]
tests/test_equivalence_partitioning.py::test_invalid_recipient_is_atomic[EP-R2] PASSED [ 88%]
tests/test_equivalence_partitioning.py::test_invalid_recipient_is_atomic[EP-R3] PASSED [ 89%]
tests/test_equivalence_partitioning.py::test_invalid_recipient_is_atomic[EP-R4] PASSED [ 90%]
tests/test_state_transitions.py::test_active_to_suspended_and_recovered PASSED [ 91%]
tests/test_state_transitions.py::test_frozen_to_active PASSED [ 92%]
tests/test_state_transitions.py::test_close_from_every_state_is_permanent[ST3] PASSED [ 92%]
tests/test_state_transitions.py::test_close_from_every_state_is_permanent[ST4] PASSED [ 93%]
tests/test_state_transitions.py::test_close_from_every_state_is_permanent[ST5] PASSED [ 94%]
tests/test_state_transitions.py::test_close_from_every_state_is_permanent[ST6] PASSED [ 95%]
tests/test_state_transitions.py::test_frozen_blocks_all_financial_operations PASSED [ 96%]
tests/test_state_transitions.py::test_insufficient_fee_suspends_then_deposit_recovers PASSED [ 96%]
tests/test_state_transitions.py::test_partial_deposit_keeps_suspended PASSED [ 97%]
tests/test_state_transitions.py::test_invalid_deposit_preserves_state[ST10] PASSED [ 98%]
tests/test_state_transitions.py::test_invalid_deposit_preserves_state[ST11] PASSED [ 99%]
tests/test_state_transitions.py::test_invalid_deposit_preserves_state[ST12] PASSED [100%]

- generated xml file: C:\Users\david\MYSHIT\homework-4\students\davidvaldez\homework-4\reports\results.xml -
===== tests coverage =====
_____ coverage: platform win32, python 3.12.14-final-0 _____

Name                               Stmts  Miss Branch BrPart  Cover   Missing
-----
src\__init__.py                     0      0      0      0  100%
src\banking_system.py              150      0     52      0  100%
-----
TOTAL                             150      0     52      0  100%
Coverage HTML written to dir htmlcov
Coverage JSON written to file reports/coverage.json
===== 125 passed in 0.86s =====
```

La captura anterior muestra una vista HTML del texto exacto del registro de terminal, no una sesión interactiva simulada. El archivo de texto enlazado contiene los 125 resultados. La siguiente captura corresponde al reporte HTML real producido por `coverage.py`.

Coverage report: 100%

Files

Functions

Classes

coverage.py v7.16.1, created at 2026-09-23 10:32 -0600

filter...

☐ hide covered

File ▴	Statements				Branches			Total
	coverage	statements	missing	excluded	coverage	branches	partial	
src__init__.py	100%	0	0	0	100%	0	0	100%
src\banking_system.py	100%	150	0	0	100%	52	0	100%
Total	100%	150	0	0	100%	52	0	100%

coverage.py v7.16.1, created at 2026-09-23 10:32 -0600

Experimento M3: exención inclusiva incorrecta

```
.....F..... [100%]
===== FAILURES =====
_____ test_savings_fee_waiver_boundary[BV-F2] _____
tests\test_boundary_values.py:67: in test_savings_fee_waiver_boundary
    assert bank.process_monthly_fee() == Decimal(fee)
E   AssertionError: assert Decimal('0') == Decimal('5')
E   +   where Decimal('0') = process_monthly_fee()
E   +   where process_monthly_fee = <banking_system.BankAccount object at 0x00000276838FE390>.process_monthly_fee
E   +   and   Decimal('5') = Decimal('5')
===== short test summary info =====
FAILED tests\test_boundary_values.py::test_savings_fee_waiver_boundary[BV-F2]
1 failed, 36 passed in 0.20s
```

La tercera imagen es evidencia del experimento, no un defecto pendiente. La solución consiste en mantener la condición original estricta `balance > threshold`; la ejecución final confirma el resultado correcto.

Interpretación y límites

EP cubre validaciones de entrada y resultados de consulta; BVA concentra su aporte en igualdades, centavos y calendario. DT añade combinaciones de restricciones, cobros y pagos futuros; ST prueba secuencias, estados terminales y operaciones prohibidas. La unión cubre todas las líneas y ramas instrumentadas. No hay líneas ni ramas marcadas como faltantes en la ejecución final.

Persisten huecos semánticos: no se enumeran todas las secuencias de eventos ni todos los pares de condiciones; faltan límites dedicados de Premium, cambio de año, tamaños extremos, zonas horarias y calendarios complejos. Los requisitos de autenticación, red, persistencia, atomicidad concurrente y liquidación externa están fuera del modelo. La cobertura no convierte esos huecos en casos probados. El clock real por defecto tampoco se valida temporalmente con esta suite determinista.

Resultado individual por técnica

equivalence_partitioning

- PASS test_transfer_amount_partition[EP-A1]
- PASS test_transfer_amount_partition[EP-A2]
- PASS test_transfer_amount_partition[EP-A3]
- PASS test_transfer_amount_partition[EP-A4]
- PASS test_transfer_amount_partition[EP-A5]
- PASS test_transfer_amount_partition[EP-A6]
- PASS test_transfer_amount_partition[EP-A7]
- PASS test_transfer_amount_partition[EP-A8]
- PASS test_transfer_amount_partition[EP-A9]
- PASS test_transfer_amount_partition[EP-A10]
- PASS test_account_type_partition[EP-T1]
- PASS test_account_type_partition[EP-T2]
- PASS test_account_type_partition[EP-T3]
- PASS test_account_type_partition[EP-T4]
- PASS test_initial_balance_partition[EP-B1]
- PASS test_initial_balance_partition[EP-B2]
- PASS test_initial_balance_partition[EP-B3]
- PASS test_initial_balance_partition[EP-B4]
- PASS test_payee_partition[EP-P1]
- PASS test_payee_partition[EP-P2]

- PASS test_payee_partition[EP-P3]
- PASS test_payee_partition[EP-P4]
- PASS test_payee_partition[EP-P5]
- PASS test_payee_partition[EP-P6]
- PASS test_history_date_partition[EP-D1]
- PASS test_history_date_partition[EP-D2]
- PASS test_history_date_partition[EP-D3]
- PASS test_history_date_partition[EP-D4]
- PASS test_history_date_partition[EP-D5]
- PASS test_history_date_partition[EP-D6]
- PASS test_history_date_partition[EP-D7]
- PASS test_update_information_partition[EP-I1]
- PASS test_update_information_partition[EP-I2]
- PASS test_update_information_partition[EP-I3]
- PASS test_transfer_between_own_accounts
- PASS test_invalid_recipient_is_atomic[EP-R2]
- PASS test_invalid_recipient_is_atomic[EP-R3]
- PASS test_invalid_recipient_is_atomic[EP-R4]

boundary_values

- PASS test_transfer_limit_boundaries[BV-A1]
- PASS test_transfer_limit_boundaries[BV-A2]
- PASS test_transfer_limit_boundaries[BV-A3]
- PASS test_transfer_limit_boundaries[BV-A4]
- PASS test_transfer_limit_boundaries[BV-A5]
- PASS test_transfer_limit_boundaries[BV-A6]
- PASS test_transfer_limit_boundaries[BV-A7]
- PASS test_transfer_limit_boundaries[BV-A8]
- PASS test_transfer_limit_boundaries[BV-A9]
- PASS test_transfer_limit_boundaries[BV-A10]
- PASS test_transfer_limit_boundaries[BV-A11]
- PASS test_transfer_limit_boundaries[BV-A12]
- PASS test_savings_minimum_boundary[BV-M1]
- PASS test_savings_minimum_boundary[BV-M2]
- PASS test_savings_minimum_boundary[BV-M3]
- PASS test_savings_minimum_boundary[BV-M4]
- PASS test_savings_minimum_boundary[BV-M5]
- PASS test_savings_fee_waiver_boundary[BV-F1]
- PASS test_savings_fee_waiver_boundary[BV-F2]
- PASS test_savings_fee_waiver_boundary[BV-F3]
- PASS test_savings_fee_waiver_boundary[BV-F4]
- PASS test_cumulative_daily_limit[BV-C1]
- PASS test_cumulative_daily_limit[BV-C2]
- PASS test_cumulative_daily_limit[BV-C3]
- PASS test_cumulative_daily_limit[BV-C4]
- PASS test_available_funds_boundary[BV-B1]
- PASS test_available_funds_boundary[BV-B2]
- PASS test_available_funds_boundary[BV-B3]
- PASS test_available_funds_boundary[BV-B4]
- PASS test_midnight_reset_boundary[BV-N1]

- PASS test_midnight_reset_boundary[BV-N2]
- PASS test_midnight_reset_boundary[BV-N3]
- PASS test_midnight_reset_boundary[BV-N4]
- PASS test_payment_date_boundary[BV-P1]
- PASS test_payment_date_boundary[BV-P2]
- PASS test_payment_date_boundary[BV-P3]
- PASS test_payment_date_boundary[BV-P4]

decision_tables

- PASS test_transfer_decision_rules[DT-T1]
- PASS test_transfer_decision_rules[DT-T2]
- PASS test_transfer_decision_rules[DT-T3]
- PASS test_transfer_decision_rules[DT-T4]
- PASS test_transfer_decision_rules[DT-T5]
- PASS test_transfer_decision_rules[DT-T6]
- PASS test_transfer_decision_rules[DT-T7]
- PASS test_transfer_decision_rules[DT-T8]
- PASS test_monthly_fee_decision_rules[DT-F1]
- PASS test_monthly_fee_decision_rules[DT-F2]
- PASS test_monthly_fee_decision_rules[DT-F3]
- PASS test_monthly_fee_decision_rules[DT-F4]
- PASS test_monthly_fee_decision_rules[DT-F5]
- PASS test_monthly_fee_decision_rules[DT-F6]
- PASS test_monthly_fee_decision_rules[DT-F7]
- PASS test_monthly_fee_decision_rules[DT-F8]
- PASS test_monthly_fee_decision_rules[DT-F9]
- PASS test_monthly_fee_decision_rules[DT-F10]
- PASS test_bill_payment_decision_rules[DT-P1]
- PASS test_bill_payment_decision_rules[DT-P2]
- PASS test_bill_payment_decision_rules[DT-P3]
- PASS test_bill_payment_decision_rules[DT-P4]
- PASS test_bill_payment_decision_rules[DT-P5]
- PASS test_bill_payment_decision_rules[DT-P6]
- PASS test_bill_payment_decision_rules[DT-P7]
- PASS test_bill_payment_decision_rules[DT-P8]
- PASS test_bill_payment_decision_rules[DT-P9]
- PASS test_bill_payment_decision_rules[DT-P10]
- PASS test_bill_payment_decision_rules[DT-P11]
- PASS test_bill_payment_decision_rules[DT-P12]
- PASS test_bill_payment_decision_rules[DT-P13]
- PASS test_bill_payment_decision_rules[DT-P14]
- PASS test_bill_payment_decision_rules[DT-P15]
- PASS test_bill_payment_decision_rules[DT-P16]
- PASS test_future_payment_processed_once
- PASS test_scheduled_payment_revalidation[DT-S2]
- PASS test_scheduled_payment_revalidation[DT-S3]
- PASS test_scheduled_payment_revalidation[DT-S4]

state_transitions

- PASS test_active_to_suspended_and_recovered

- PASS test_frozen_to_active
- PASS test_close_from_every_state_is_permanent[ST3]
- PASS test_close_from_every_state_is_permanent[ST4]
- PASS test_close_from_every_state_is_permanent[ST5]
- PASS test_close_from_every_state_is_permanent[ST6]
- PASS test_frozen_blocks_all_financial_operations
- PASS test_insufficient_fee_suspends_then_deposit_recovers
- PASS test_partial_deposit_keeps_suspended
- PASS test_invalid_deposit_preserves_state[ST10]
- PASS test_invalid_deposit_preserves_state[ST11]
- PASS test_invalid_deposit_preserves_state[ST12]